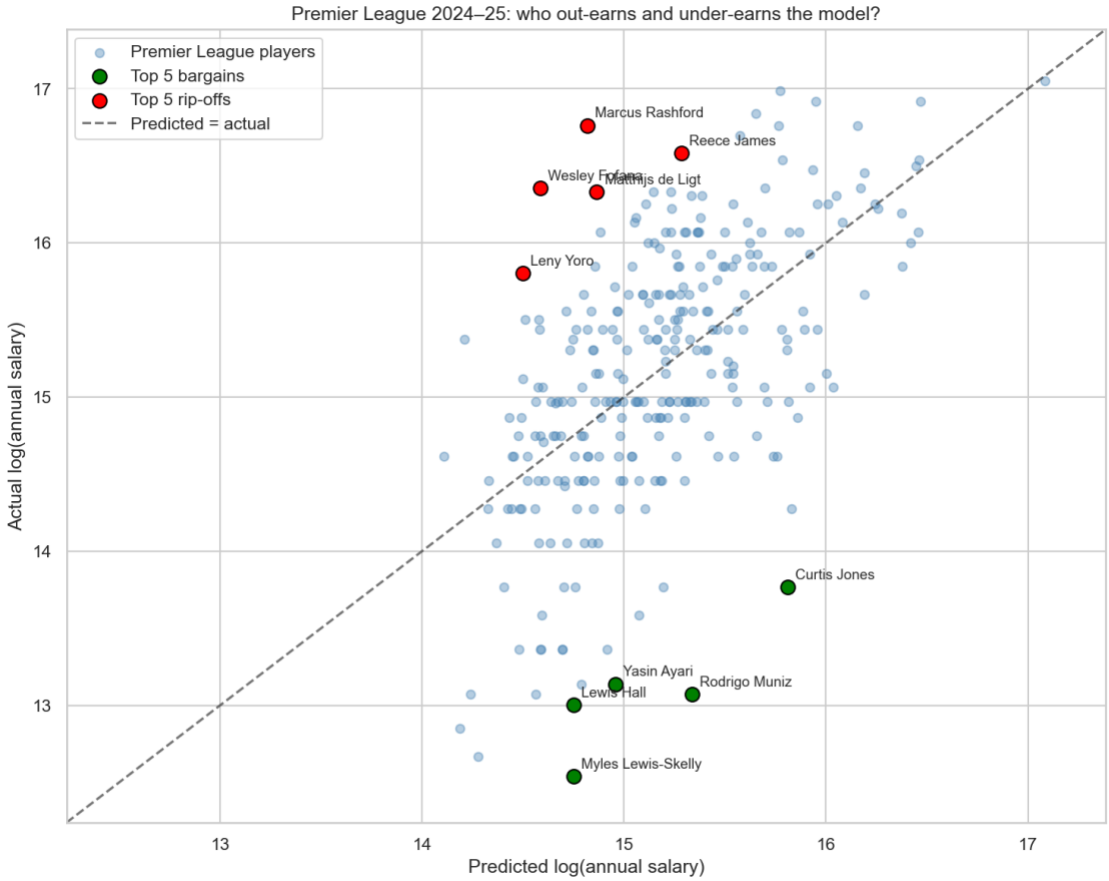
Premier League Bargains & Rip-offs of 2024–25

A data-driven look at which Premier League players over- and under-deliver on their salary.

Executive Summary

Premier League salaries are shaped by a long list of factors no public dataset can fully see: commercial appeal, agent leverage, contract timing, and how badly a club wanted a player on the day they signed. Using one season of public data on individual performance, team performance, and salaries, we built a model that explains roughly a third of what drives a player's annual pay from on-pitch contribution alone. The most interesting finding isn't the model itself; it's the players whose actual pay falls furthest from what the model predicts.

- **The biggest bargains of 2024–25 are Rodrigo Muniz and Myles Lewis-Skelly.** Fulham's backup striker (24, 964 minutes) earned roughly **£476k** against a model prediction of ≈ **£4.6 million**. Arsenal's 18-year-old academy defender (1,348 minutes) earned just **£279k** against a model prediction of ≈ **£2.5 million**. Both are senior contributors on contracts that haven't caught up with their on-pitch output.
- **The biggest rip-off is Marcus Rashford.** He earned roughly **£19.1 million** against a model prediction of about **£2.7 million**, a gap so wide it captures exactly why Manchester United loaned him out mid-season.
- **Manchester United and Chelsea each have three players in the top ten rip-offs:** between them, six of the ten most over-paid players in the league. The list reads like a leaderboard of contracts that aged poorly.
- **Even Erling Haaland**, the league's most prolific scorer, appears on the rip-off list. Not because he's a bad player, but because his ≈ £23.8M wage is roughly **3.4× higher** than the model's prediction for someone with his on-pitch profile. A reminder that "rip-off" here means over-paid relative to comparable performers, not bad at football.
- Beneath the headline names, the model also confirms a more subtle finding: knowing what fraction of their team's output a player contributes, not just their raw stat totals, meaningfully improves how well we can predict their salary. The technical version of that statement is $F(5, 274) = 7.82$, $p \approx 6.7 \times 10^{-7}$; the plain-English version is the chance that pattern is random noise is roughly one in 1.5 million.



Predicted vs. actual salary

Each dot is one Premier League player. Players on the dashed line earn roughly what the model predicts. Green dots earn far less than predicted (bargains); red dots earn far more (rip-offs).

The Data

We combined three datasets covering the 2024–25 Premier League season, downloaded from a public Kaggle bundle that aggregates two independent upstream collections. The first is a **player performance** dataset (571 player-season rows) with each player's season totals: goals, assists, expected goals, progressive passes and carries, minutes, age, and position. The second is a **player salaries** dataset (684 rows) with each player's weekly and annual gross salary in GBP. The third is a **team performance** dataset (20 rows) with each club's season totals across the same on-pitch categories, plus possession.

Although all three CSVs are distributed together on Kaggle, the **performance** data and the **salary** data come from two genuinely different upstream collections produced by independent organizations. The on-pitch statistics are scraped from match-event data and aggregated into season totals; the salary figures are compiled separately from publicly reported and estimated player wages. The two collections cover overlapping but not identical sets of players (571 player-seasons in the stats table; 684 in the salaries table), and the stats table contains duplicate

rows for players who transferred between Premier League clubs mid-season. Both had to be handled deliberately during the merge.

Cleaning and Merging

A handful of players appear more than once in the performance data because they transferred between Premier League clubs mid-season. We collapsed these duplicates by keeping each player's row at the team where they played the most minutes, treating that as their "primary" club for the season. To guard against the rare case where two genuinely different players share a name, we identified each player by both their name and their year of birth, not just their name. As a safety net, the merge step itself was configured to halt with an error if any duplicate name had slipped through; it never triggered.

The two merges produced a clean player-level dataset of **514 rows × 34 columns**, with each row combining one player's individual stats, their salary, and their team's aggregate stats.

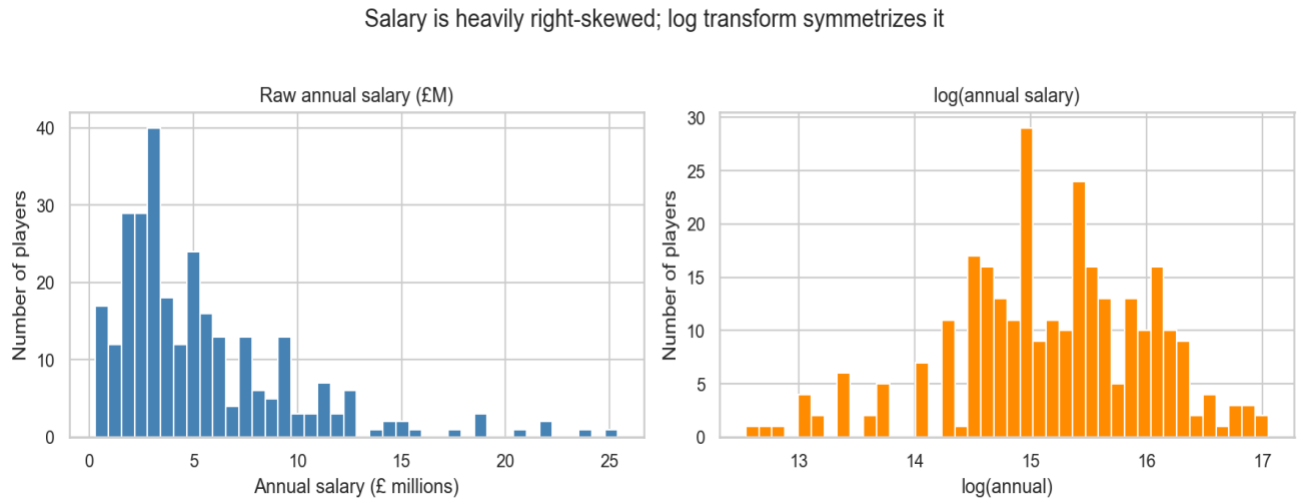
After merging, we filtered the player population for the analysis:

- **Minimum 900 minutes played** (≈ 10 full matches, the standard threshold in football analytics for treating per-match rates as stable rather than small-sample noise). This filter removes **201 players**.
- **Goalkeepers excluded** (a further **25 players**). Our dataset contains no goalkeeper-specific statistics (saves, clean sheets, post-shot expected goals against), so the model cannot fairly evaluate them on outfield-style contribution metrics. We acknowledge this as a limitation below.

The analysis sample is **288 outfielders** with at least 900 minutes: the population of "regular starters and trusted rotation players" in the 2024–25 Premier League.

How we measured "value for money"

We used **multiple linear regression**, a standard tool from labor economics for asking "how is a person's pay related to what they measurably contribute?" The model is trained to predict salary on the logarithmic scale rather than in raw pounds. This is a routine adjustment for any kind of income data: it pulls in the long tail of mega-earners so a handful of superstars don't dominate the fitting process, and it makes the model's predictions multiplicative, so coefficients can be read as approximate percentage changes in pay rather than absolute pound amounts.



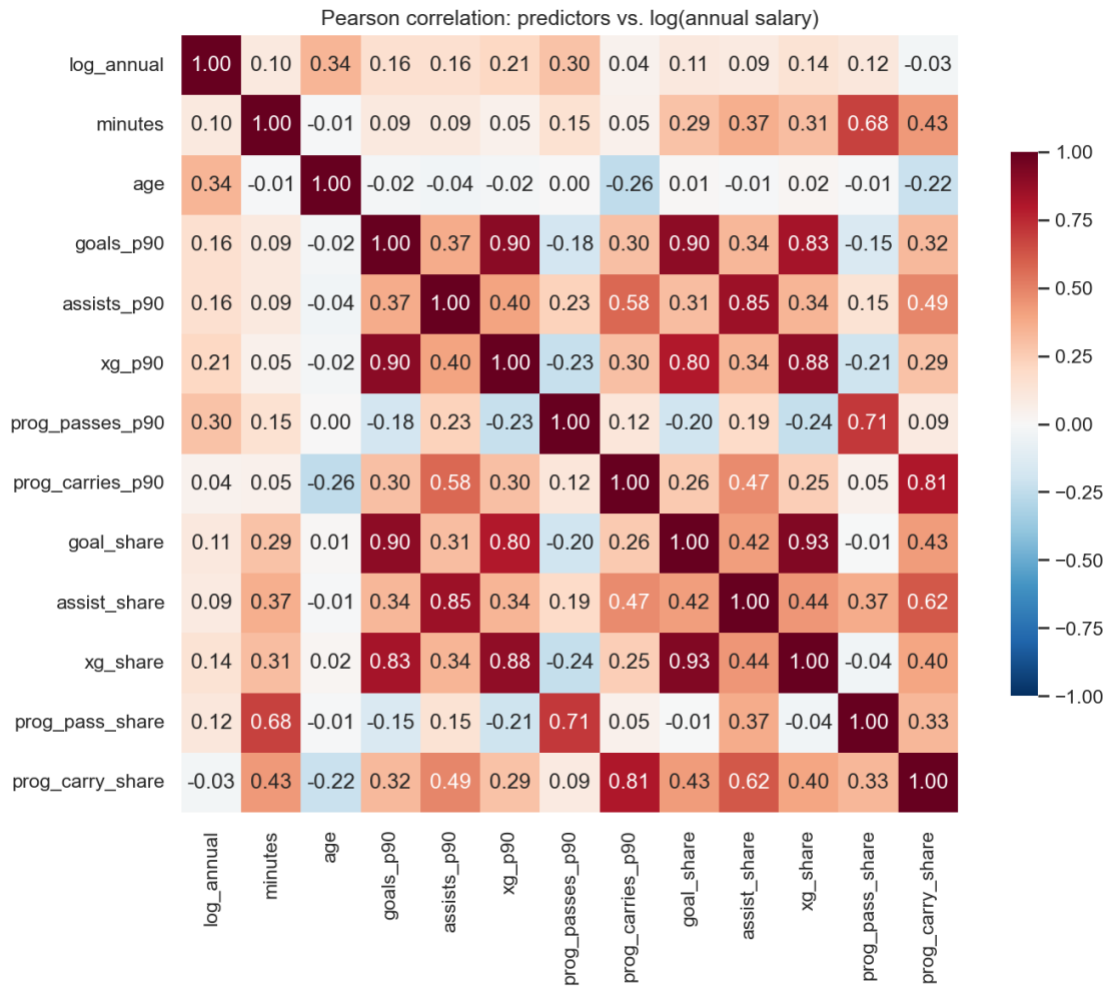
Salary distribution before and after log transform

Left: raw salaries are heavily skewed, with most players bunched at the low end and a thin tail of stars stretching out to the right. Right: the same salaries on the logarithmic scale form a roughly bell-shaped distribution, which is what the regression needs to fit reliably.

The predictors fall into three groups:

1. **Playing-time and demographics.** Minutes played, age, primary position (defender, midfielder, or forward).
2. **Per-90 performance rates.** Goals, assists, progressive passes, and progressive carries, each expressed as a rate per 90 minutes played rather than as a season total. This is the standard adjustment in football analytics: it ensures we compare a player who played 1,000 minutes and a player who played 3,000 minutes on the same scale, by asking "how productive were they per match's worth of time on the pitch?"
3. **Share of team output.** What fraction of their team's goals, assists, expected goals, progressive passes and progressive carries did this player contribute? These are the variables that make the team-level data essential; without it, "share of team" is undefined.

Before fitting the model, we examined how each predictor correlates with log salary:



Correlation heatmap

A statistical measure of how strongly each predictor moves with salary, ranging from −1 to +1. Age, progressive passes, expected goals, and assists are the strongest individual signals. The share-of-team measures carry distinct information from their raw counterparts: they're related, but not so closely as to be redundant.

The Hypothesis Test

The most important question of the project: **do the share-of-team measures explain salary variation that raw performance alone does not?** Answering it also justifies why we needed the team-level data at all.

We compared two versions of the model. The **baseline** uses only individual stats (minutes, age, position, and the per-90 rates of goals, assists, progressive passes, and progressive carries) and explains **23.2%** of the variation in log-salary. The **full model** adds the five share-of-team measures on top of the baseline, and lifts the explained variation to **31.6%**, an increase of roughly 8 percentage points.

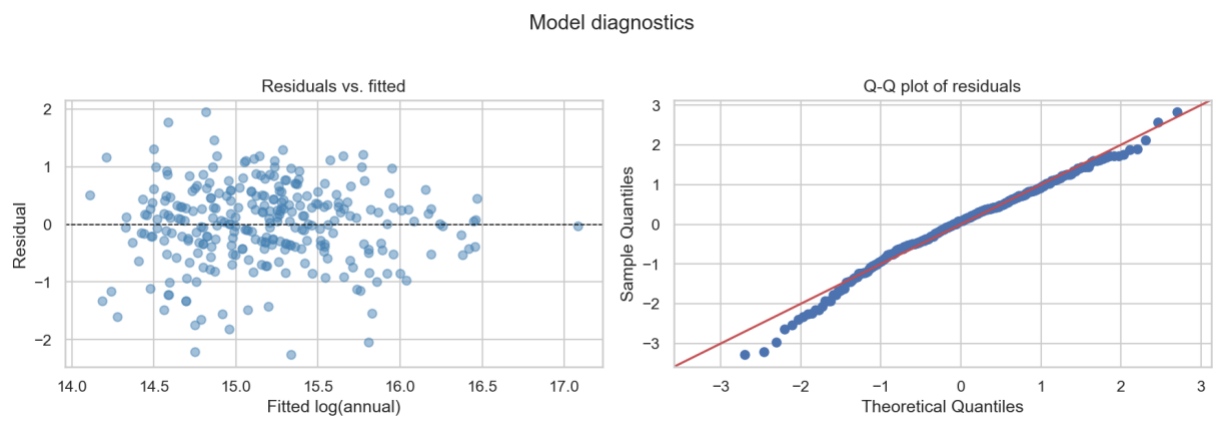
The question is whether that improvement is large enough to be a real effect, or whether it could plausibly be random noise. The standard test for this comparison gave **$F(5, 274) = 7.82$, $p \approx 6.7 \times 10^{-7}$**. In plain English: the chance that the share-of-team measures would improve the model this much by random luck is essentially zero, about one in 1.5 million. **The team-level data is doing real work**, and the full model is the basis for the bargain and rip-off rankings below.

Results

The model fits salary moderately well

The full model explains **31.6%** of what drives Premier League salaries (adjusted R^2, a standard goodness-of-fit score that ranges from 0% to 100%). For a one-season pay model with no information beyond on-pitch contribution, that's a respectable fit. The remaining two-thirds is explained by factors we don't observe in the data: the player's commercial appeal, the length and timing of their contract, agent leverage, and how desperately the buying club needed the position the day they signed.

Diagnostics check out



Residual diagnostics

Two standard checks that the model isn't being misleading. **Left:** the model's prediction errors are spread roughly evenly above and below zero across the entire predicted-salary range, with no obvious pattern, which is what we want to see. **Right:** the same prediction errors compared against a normal bell-curve; they line up well in the middle and only deviate at the extremes, which is the expected pattern for salary data because a handful of superstars sit further out than a normal curve would predict. The model's standard statistical assumptions are reasonably met.

Top 10 bargains of 2024–25

The 10 players whose actual salary falls furthest below what the model predicts. Numbers in £.

Rank	Player	Team	Pos	Age	Minutes	Actual	Predicted
1	**Rodrigo Muniz**	**Fulham**	**FW**	**24**	**964**	**£476,288**	**£4,575,410**
2	**Myles Lewis-Skelly**	**Arsenal**	**DF**	**18**	**1,348**	**£279,230**	**£2,547,000**
3	Curtis Jones	Liverpool	MF	24	1,621	£952,577	£7,339,021
4	Yasin Ayari	Brighton	MF	21	1,921	£508,041	£3,133,420
5	Lewis Hall	Newcastle Utd	DF	20	2,189	£444,536	£2,544,337
6	Cameron Burgess	Ipswich Town	DF	29	1,541	£508,041	£2,646,873
7	Tyler Dibling	Southampton	FW	19	1,874	£317,526	£1,586,473
8	Kevin Schade	Brentford	FW	23	2,214	£635,051	£3,005,486
9	Yoane Wissa	Brentford	FW	28	2,829	£1,587,628	£7,479,423
10	Carlos Baleba	Brighton	MF	21	2,571	£793,814	£3,518,612

Pattern: The bargain list is dominated by two profiles. First, **young academy graduates and emerging starters** still on early-career contracts (Lewis-Skelly, Dibling, Baleba, Hall): clubs cannot easily pay them market rate because of how youth and rookie contracts are structured. Second, **prolific contributors at smaller or mid-table clubs** whose wage bills sit well below the elite tier (Muniz at Fulham, Wissa and Schade at Brentford, Burgess at Ipswich): they are putting up numbers that would command Big-Six salaries if they were at a Big-Six club. **Curtis Jones (Liverpool)** is the most striking outlier on the list: an established starter at a top club who, on this model, looks priced more like a depth player than a regular contributor to a title-winning side.

Top 10 rip-offs of 2024–25

The 10 players whose actual salary falls furthest above what the model predicts. Numbers in £.

Rank	Player	Team	Pos	Age	Minutes	Actual	Predicted
1	**Marcus Rashford**	**Manchester Utd**	**FW**	**27**	**978**	**£19,051,534**	**£2,723,392**
2	Wesley Fofana	Chelsea	DF	24	1,172	£12,701,023	£2,157,891
3	Matthijs de Ligt	Manchester Utd	DF	25	2,123	£12,383,497	£2,856,517
4	Leny Yoro	Manchester Utd	DF	19	1,165	£7,303,088	£1,980,458
5	Reece James	Chelsea	DF	25	973	£15,876,279	£4,338,995
6	Erling Haaland	Manchester City	FW	24	2,652	£23,814,418	£7,084,663
7	Omar Marmoush	Manchester City	FW	26	1,094	£20,593,224	£6,268,796
8	Christopher Nkunku	Chelsea	FW	27	921	£12,383,497	£3,774,897
9	Jarrod Bowen	West Ham	FW	28	2,883	£9,525,767	£2,910,289
10	Archie Gray	Tottenham	DF	19	1,661	£4,762,884	£1,484,033

Pattern: Manchester United and Chelsea each have three players in the top ten, between them six of the league's ten most over-paid players. The list reads like a leaderboard of contracts that aged poorly: high-wage signings whose 2024–25 minutes or output didn't match what they were earning when the deal was agreed. Several entries reflect injury-shortened seasons (Reece James, Wesley Fofana) or players frozen out by their managers (Rashford, Nkunku); others were simply expensive transfers paying a premium the model can't price (Haaland, Marmoush, de Ligt). **Erling Haaland's** appearance on this list is the model's most counter-intuitive output: he is the most prolific scorer in the league, and yet his wage is roughly **3.4× higher** than what the model predicts for someone with his profile. The model is not saying he's a bad signing; it is saying his contract sits at a tier of its own, and very few players in the dataset earn enough to be benchmarks against him.

Limitations and honest caveats

- **One season only.** This is a 2024–25 snapshot. A player who looks under-paid here may be on a contract signed years ago when they were less productive; the data can't see that history.

- **Goalkeepers were excluded.** Our dataset has no goalkeeper-specific statistics, so we cannot fairly score keepers. The bargain/rip-off rankings cover outfield players only.
- **Salary data is third-party estimated.** Premier League clubs do not publish exact wages, so the salary dataset is compiled from publicly reported and estimated figures. Treat the rankings as directionally informative, not penny-precise.
- **The model only sees on-pitch contribution.** Real-world salaries also reflect commercial value (shirt sales, sponsorships), age and contract leverage at signing, and squad-needs at the time of the deal. Players we flag as "rip-offs" may be perfectly justified on dimensions our data doesn't capture. For instance, an injury-shortened season for an otherwise productive player will look like a rip-off in our data without being one.
- **Position is collapsed to three categories.** A defensive midfielder and an attacking midfielder both fall under "MF," but their pay structures differ.
- **Some predictors point the "wrong" way.** The model assigns a negative weight to several share-of-team measures, most strongly a player's share of their team's progressive passes, implying that, holding per-90 output and minutes constant, taking on a larger fraction of a team's build-up play is associated with lower pay. The most plausible explanation: at smaller and weaker clubs, one or two players carry an outsized share of the build-up burden, and those clubs pay below the league-wide rate. The share-of-team measure is partially picking up "this is a small-club player." A future version of the model could control directly for team quality (for example, the team's overall expected goals or possession share) to disentangle the two effects.

Conclusion

A simple regression on one season of public data identifies precisely the kinds of players football executives spend their careers thinking about: **academy graduates whose contracts haven't caught up to their performance, and high-wage signings whose deals have aged badly**. The model is not a verdict on any individual player's quality; Marcus Rashford is a vastly better footballer than most of the league, and the model knows that. It is a verdict on the gap between what a player produces and what their club pays them, given how the rest of the league is currently priced.

If you ran a Premier League recruitment desk, you'd use a model like this not to settle arguments but to start them. The bargain list highlights players whose current contracts will become problems for their clubs in the next renewal window. The rip-off list highlights signings that won't repeat and helps frame which positions to avoid bidding up in the next transfer market. One season of data is enough to surface those candidates; multiple seasons would let you separate genuine value from short-term variance.